Royal Devon University Healthcare NHS Foundation Trust

Template: October 2024

Guidance for the perioperative management of **Steroid replacement in adult patients with adrenal insufficiency**

Author of Clinical Guideline	*Dr Anna Perham, Consultant Anaesthetist* *Dr Naomi Adey, Core Anaesthetic Trainee*
Division/ Department responsible for Clinical Guideline	*Surgical Services, Preparation for Surgery and Anaesthesia*
Contact details	*Anna.perham@nhs.net*
Are there any previous pieces of guidance relating to this topic?	Yes
Title of previous versions if applicable so it can be archived once this document is approved.	Guidelines for peri-operative steroid replacement in patients with adrenal insufficiency
Version number	1.0
Replaces version number	
Date written	*25/03/25*
Consultation undertaken with:	*Dr Antonia Brooke and Dr Roderick Warren, Endocrinology Consultants* *Endocrine Specialist Nurses,* *Dr Kat Haynes, Consultant in Anaesthesia and Perioperative medicine,* *Alison Lamb, specialist pharmacist*
Approving body and date approved	*08/07/2025*
Review date	*25/09/2029*
Expiry date	*25/03/2030*
Date document becomes live	*17/07/2025*
Search terms for the Hub include abbreviations, short phrases, common mis-spellings or alternative spellings e.g. oedema, edema	*Adrenal Insufficiency* *Adrenal Crisis* *Steroids* *Steroid replacement* *Prednisolone* *Hydrocortisone*

CONTENTS

1. INTRODUCTION 3
2. BACKGROUND 3
3. DEFINITIONS 3
4. MAIN BODY OF GUIDELINE 4
5. MONITORING COMPLIANCE WITH THIS GUIDELINE 9
6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES 9

1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 These guidelines aim to ensure that patients with adrenal insufficiency are adequately supplemented with glucocorticoids in the peri-operative period.

The reference texts for this document are:

- NICE NG243 – "*Adrenal insufficiency: identification and management*" (2024)
- UK consensus – "*Guidelines for the management of glucocorticoids during the peri-operative period for patients with adrenal insufficiency*" (2020) – at https://doi.org/10.1111/anae.14963
- European Society of Endocrinology and Endocrine Society Joint Clinical Guideline – "*Diagnosis and therapy of glucocorticoid-induced adrenal insufficiency"* (2024) – at https://doi.org/10.1093/ejendo/lvae029
- Society for Endocrinology patient information leaflet – "*Steroid sick day rules*" at https://www.endocrinology.org/media/4142/ai-and-exogenous-steroids_pis_final.pdf

2. BACKGROUND

2.1 The need for an RDUH guideline arises because the above guidelines provide slightly varying advice, and in some instances advice which is complex or counter-intuitive and without a clear evidence base. This guideline represents the opinion of the RDUH endocrinology consultant body and both the northern and eastern anaesthetic departments.

3. DEFINITIONS

3.1 "Glucocorticoids" is the most accurate term for drugs such as prednisolone, hydrocortisone and dexamethasone. "Steroids" technically also encompasses other drug classes. However, for brevity and to reflect everyday clinical use, henceforth this guideline will use "steroid" in place of "glucocorticoid".

4. GUIDANCE FOR THE MANAGEMENT OF STEROID REPLACEMENT

Clinical principles

Patients with adrenal insufficiency are at risk of adrenal crisis and must be given stress doses of steroids during the peri-operative period due to the physiological stress response.

Patients who already have a diagnosis of adrenal insufficiency should already be treated with physiological steroid replacement.

Many patients receive long-term steroids for other medical conditions. This is the most common cause of adrenal suppression. These patients are most likely to have undiagnosed adrenal insufficiency, and are therefore at risk of omitted or inadequate steroid treatment during the perioperative period.

Identification of patients at risk of undiagnosed adrenal insufficiency due to exogenous steroid treatment

There are no firm guidelines regarding the dose or duration of steroid treatment that should warrant concern for adrenal insufficiency.

NICE NG243 advises that adrenal insufficiency is "*more common*" in adults who have recently stopped using steroids by any route of administration after taking them for more than 4 weeks, or are currently taking a physiological replacement dose of steroids. A physiological replacement dose is defined as 15-25 mg oral hydrocortisone, 3-5 mg prednisolone, 0.5 mg dexamethasone.

The European Societies joint guideline says that "*Glucocorticoid exposure via oral administration that poses risk for adrenal insufficiency, is expected to at least exceed both of the following thresholds:*

- *Duration of 3-4 weeks, or greater*
- *Dose greater than daily hydrocortisone equivalent of 15-25 mg (4-6 mg prednisolone, 3-5 mg methylprednisone, 0.25-0.5 mg dexamethasone)"*

If in doubt about the need for steroid replacement, it should be given, as there are no long-term adverse consequences of short-term administration.

Any post-operative complications may result in a prolonged stress response and a requirement for ongoing steroid supplementation. This should be based on clinical judgement.

Route of steroid replacement in the peri-operative period

Oral steroids should be replaced by IV/IM steroids in the peri-operative period. The suggested steroid regimen for patient groups and types of surgery is described in Tables 1 and 2 below. Continuous IV infusion is preferred in published reference guidelines, and may be suitable in high dependency settings. Intermittent IV dosing is favoured in this guideline to reduce the need for a dedicated cannula and infusion pump, and may be safer if there is a risk of losing venous access. However, Hydrocortisone 200 mg/24 hours by continuous IV infusion is an alternative to Hydrocortisone 50 mg IV QDS in high dependency settings. Hydrocortisone 50 mg QDS by intramuscular injection is another acceptable option. Doses

of IV Hydrocortisone are medications that should not be omitted and must be given IM if IV access is temporarily lost.

Explanatory note about dexamethasone

Dexamethasone is a pure glucocorticoid. It has no mineralocorticoid activity. This means it is suitable replacement steroid for patients who have adrenal suppression due to steroid treatment, and for patients who have adrenal insufficiency due to pituitary disease – in both scenarios, patients should have an intact mineralocorticoid axis.

Dexamethasone is not suitable for patients with primary adrenal failure (Addison's disease) who also need mineralocorticoid replacement. These patients typically take oral fludrocortisone. However, hydrocortisone is a partial mineralocorticoid, and in the doses used during surgery or illness, IV hydrocortisone is sufficient.

Our view is that the patients at greatest risk are those with current or recent steroid treatment, in whom the possibility of adrenal suppression may be overlooked. We also note that it may not always be obvious whether a patient has primary adrenal or pituitary disease. For simplicity, patients are divided into those who have adrenal or pituitary disease (table 1) and those who have had steroid treatment (table 2).

Cases under local anaesthesia

These are outside the scope of this guideline as many of these will be done without the presence of an anaesthetist. Patients may have been advised to follow guidance from endocrinology. If the physiological insult from surgery is minor then perioperative steroid replacement may not be warranted. However, it is reasonable to treat these patients in the same way was as a day case procedure under GA/RA, especially if the magnitude of the surgery is significant.

Template: October 2024

Table 1: Peri-operative management of patients with primary adrenal insufficiency (e.g. Addison's disease) or secondary insufficiency (Pituitary disease).

Note that Dexamethasone is not a suitable alternative to hydrocortisone in this group as it has no mineralocorticoid activity.

If expert help is needed for this group of patients please contact the endocrine specialist nurses vie epic instant message on **RDE endocrine CNS** or phone on **ext. 2847**

	Intraoperative steroid replacement	**Postoperative steroid replacement**
Any surgery under GA or RA	Hydrocortisone 100 mg intravenously on induction. Followed by Hydrocortisone 50mg IV/IM every 6 hours starting 6 hours after induction. **To continue until oral steroids restarted.** **Dexamethasone is not a suitable alternative to hydrocortisone in this group.**	Hydrocortisone 50 mg intravenously every 6 hours from 6 hours after induction until clinically stable and able to take oral steroids. Intramuscular is an alternative if no IV access. **Restarting oral steroids:** ***If usual maintenance dose exceeds prednisolone 10 mg daily or hydrocortisone 40 mg daily***: Prescribe usual dose and continue until discharge. Doubling doses above these thresholds is supra-physiological and unnecessary. Hydrocortisone should be prescribed in divided doses e.g. 20 mg morning, 10 mg midday and 10 mg 6pm. ***If usual maintenance dose is less than prednisolone 10 mg daily or hydrocortisone 40 mg daily***: Increase to prednisolone 10 mg OR hydrocortisone 20 mg morning, 10 mg midday and 10 mg 6pm. Reduce to usual maintenance dose after • 24 hours – rapid recovery / day case surgery • 48 hours – moderate surgery / inpatients / major surgery • Up to 7 days – if recovery is complicated
Bowel Procedures – requiring laxatives / enema	Bowel prep under clinical supervision. Consider IV fluids and IV/IM glucocorticoid during preparation, especially for fludrocortisone or	Resume oral/enteral steroids. **Restarting oral steroids:** ***If usual maintenance dose exceeds prednisolone 10 mg daily or hydrocortisone 40 mg daily***:

	vasopressin-dependent patients. Please discuss with endocrine specialist nurses if any concerns. Hydrocortisone 100 mg intravenously or intramuscularly at start of procedure.	Prescribe usual dose and continue until discharge. Doubling doses above these thresholds is supra-physiological and unnecessary. Hydrocortisone should be prescribed in divided doses e.g. 20 mg morning, 10 mg midday and 10 mg 6pm. ***If usual maintenance dose is less than prednisolone 10 mg daily or hydrocortisone 40 mg daily***: Increase to prednisolone 10 mg OR hydrocortisone 20 mg morning, 10 mg midday and 10 mg 6pm. Reduce to usual maintenance dose after 24 hours.

Table 2: Peri-operative management of patients with adrenal insufficiency due to tertiary insufficiency (suppression from previous steroid treatment); those on oral steroids at the time of surgery, or those considered to be at high risk of deficiency.

If aetiology of adrenal insufficiency is unclear, follow guideline for primary adrenal insufficiency (Table 1)

	Intraoperative steroid replacement	**Postoperative steroid replacement**
Any surgery under GA or RA	Hydrocortisone 100 mg intravenously on induction. Followed by Hydrocortisone 50mg IV/IM every 6 hours starting 6 hours after induction. **To continue until oral steroids restarted.** ***Alternatively:*** Dexamethasone 6.6mg intravenously, if used at induction, will suffice for 24 hours.	Hydrocortisone 50 mg IV/IM every six hours until clinically stable and able to take oral medications. **If not on oral steroids at the time of surgery (but considered high risk of deficiency so given intraoperative cover), there is no need to start steroids post operatively.** **Restarting oral steroids:** ***If usual maintenance dose exceeds prednisolone 10 mg daily or hydrocortisone 40 mg daily***: Prescribe usual dose and continue until discharge. Doubling doses above these thresholds is supra-physiological and unnecessary. Hydrocortisone should be prescribed in divided doses eg. 20 mg morning and 10 mg midday and 10 mg 6pm. ***If usual maintenance dose is less than prednisolone 10 mg daily or hydrocortisone 40 mg daily***: Increase dose to prednisolone 10 mg OR hydrocortisone 20 mg morning and 10 mg midday and 10 mg 6pm. Reduce to usual maintenance dose after • 48 hours – day case / moderate surgery or major surgery with uncomplicated recovery • Up to 7 days – If recovery is complicated
Bowel Procedures – requiring laxatives / enema	Continue normal glucocorticoid dose. Equivalent IV dose if prolonged nil by mouth.	

	Treat as per primary adrenal insufficiency if concerned about HPA axis function, and risk of adrenal insufficiency	

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Patients at risk of adrenal insufficiency receive adequate steroid replacement in the peri-operative period.	*Peri-operative planning for elective cases. Appropriate prescribing and intraoperative administration of steroids.*	*Potential for future audit.*

6. REFERENCES

Woodcock et al. 'Guidelines for the management of glucocorticoids during the peri-operative period for patients with adrenal insufficiency'. Anaesthesia 2020, 75, 654–663.

NICE NG243 – "Adrenal insufficiency: identification and management" (2024).

Beuschlein et al. 'European Society of Endocrinology and Endocrine Society Joint Clinical Guideline: Diagnosis and therapy of glucocorticoid-induced adrenal insufficiency'. European Journal of Endocrinology, 2024, 190, G25–G51.

Society for Endocrinology patient information leaflet – "*Steroid sick day rules*" at https://www.endocrinology.org/media/4142/ai-and-exogenous-steroids_pis_final.pdf